

DUT DataScientia Living Lab Project Concept

Projects are central to the DUT DataScientia Living Lab initiative because they provide a practical mechanism for translating Diversity-aware Sovereign AI principles into working prototypes. Each project can serve as a demonstrator of how students, researchers, communities, and industry partners can use AI, data, local infrastructure, indigenous languages, community participation, and ethical governance to address real African challenges. The following projects show how the Living Lab can connect teaching, applied research, community engagement, and innovation incubation.

Project Name: A Sovereign AI Blueprint for Carbon Footprint Hotspot Detection in KZN Computer Labs

Project Short Description: Our project is an AI-powered system that has the ability to detect energy wasted for individual computers in DUT computer labs, it then translates the technical alerts into IsiZulu (and hopefully other indigenous languages in the future) and delivers simple SMS notifications (since WhatsApp is not African owned) to lab managers. Our solution uses open-source tools (Eco2AI), African-built language AI (Vulavula API), and local cloud hosting (Cloud Africa) to ensure data sovereignty and POPIA compliance, helping reduce electricity costs and carbon emissions.

Component	Expanded description
Problem being addressed	DUT lab administrators receive complex utility data and system logs in English. While these administrators are tech-savvy, they lack electrical engineering training and, as non-native English speakers, struggle to interpret billing data and technical logs. Based on discussions with two DUT lab staff, this language-technical skills gap has led to undetected energy waste and unnecessarily high electricity costs. Existing tools like NovaVue and GreenX monitor entire buildings only (not individual computers) and provide no indigenous language interface. This is a public service failure: the data exists, but the people managing the labs cannot use it effectively. The problem is important because it contributes to higher carbon emissions, wastes university funds (estimated R50,000-R80,000 annually per lab block), and undermines South Africa's sustainability and energy efficiency goals.
Proposed solution	Our solution is a three-step sovereign AI system: (1) Detect - install the open-source Eco2AI Python framework on each lab computer to capture real-time telemetry data (CPU/GPU wattage draw, active process IDs, runtime durations, idle time); (2) Translate - use the African-built Vulavula API (Lelapa AI) to translate technical energy data and alerts into isiZulu and other South African indigenous languages; (3) Alert - deliver simple, actionable SMS notifications to lab managers (e.g., "Computer I4B is sala ivuliwe. Vala manje" / "Computer I4B was left on. Close it now"). The system provides per-computer granularity that existing tools lack, at near-zero hardware cost using existing lab computers. SMS was chosen over WhatsApp to maintain full African sovereignty, WhatsApp is not African owned, while SMS uses local African telecommunications infrastructure.
Sovereign AI contribution	The project strengthens AI sovereignty through: (1) Local infrastructure - all data hosted on Cloud Africa (African Data Centre network on South African soil), ensuring POPIA compliance and zero data leakage to foreign jurisdictions; (2) African-built language AI - integration of Vulavula API,

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	built by African researchers for African languages, providing true semantic processing; (3) Open-source foundation: using Eco2AI (modifiable to match South African municipal electricity tariffs and Eskom grid carbon intensities) without proprietary licensing fees; (4) African telecommunications: SMS delivery uses local African network infrastructure rather than foreign-owned messaging platforms; (5) Data governance: explicit consent mechanisms and "Vala/Sula" (Close/Delete) command for Right to be Forgotten; (6) Local knowledge sharing: contributing all code to GitHub, Masakhane Web, and SADiLaR for the broader African AI community.
Diversity-aware AI contribution	The project embeds diversity and inclusion at every layer: (1) Language diversity - Vulavula API supports multiple South African indigenous languages (starting with isiZulu, expandable to isiXhosa, Sesotho, etc.), breaking the English-only barrier that excludes non-native speakers from understanding and acting on energy data; (2) Accessibility - SMS-based delivery works on any phone (including basic feature phones), unlike WhatsApp which requires smartphones and data; (3) Digital inclusion - reaches users regardless of smartphone ownership, data affordability, or network connectivity requirements; (4) Cultural context - language is adapted to local usage patterns and includes culturally relevant terminology; (5) Skills inclusion - reduces reliance on electrical engineering expertise by translating technical metrics into plain language; (6) Community participation - stakeholders include lab managers, ICT/IS Department, lecturers, students, Green Campus Initiative, and Green Engineering teams
Primary users and beneficiaries	Primary users: Lab administrators and managers at DUT computer labs who need to understand complex energy data to act. ICT/IS Department is responsible for infrastructure costs and will benefit from reduced electricity bills. Secondary beneficiaries: Students and lecturers who use the labs and benefit from a sustainable, efficiently managed environment. Green Campus Initiative & Renewable Energy teams: who need granular data for sustainability reporting and carbon reduction goals. Broader community: other KZN universities and educational institutions that can adopt the open-source solution. How they benefit: Lab managers receive actionable alerts in their home language delivered via SMS (accessible on any phone), reducing cognitive load and enabling quick action; ICT saves R50,000-R80,000 annually in electricity costs; students experience a greener campus; Green Campus Initiative gains data for sustainability metrics.
Expected outputs	(1) Deployed prototype on 5-10 DUT lab computers within 3 months, (2) Fully functional Eco2AI telemetry script capturing per-computer energy data, (3) Vulavula API integration layer translating technical alerts into isiZulu, (4) SMS delivery system (using local African telecommunications infrastructure), (5) Cloud Africa-hosted database of telemetry data (POPIA compliant), (6) GitHub repository with open-source codebase (telemetry scripts, API integration, SMS pipeline), (7) SADiLaR contribution of localized language data, (8) Technical report/documentation, (9) Pilot evaluation report

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	measuring energy savings and user comprehension rates, (10) Scalability roadmap for other KZN institutions.
Research potential	Our project offers significant research and innovation opportunities: (1) Publication opportunities in AI for sustainability, human-computer interaction, and African natural language processing journals/conferences; (2) Contributes to the growing body of research on indigenous language AI applications in Africa; (3) Provides a replicable model for other African institutions facing similar language-energy challenges; (4) Potential for future funding from Technology Innovation Agency (TIA) Seed Funding Scheme and Department of Science and Innovation (DSI); (5) Establishes partnerships with Lelapa AI (Vulavula) and Cloud Africa for ongoing collaboration; (6) Can scale to other universities, government buildings, and public institutions across KZN and South Africa; (7) Contributes to SDG research (SDG 7 Clean Energy, SDG 9 Innovation, SDG 13 Climate Action); (8) Creates a longitudinal dataset on computer energy usage patterns in South African higher education.
Immediate next steps	(1) Weeks 1-2: Install Eco2AI on 5 test lab computers and begin data collection; obtain ethics clearance and POPIA consent protocols from DUT; confirm stakeholder engagement with ICT/IS Department and lab managers; evaluate SMS gateway providers (local African options preferred). (2) Weeks 3-4: Develop basic Python script for Eco2AI telemetry on DUT lab workstations; integrate Vulavula API sandbox (free tier) for isiZulu translation testing; prototype SMS delivery system connection. (3) Weeks 5-8: Conduct pilot testing with 2 DUT lab staff to validate alert comprehension and usability; refine translation accuracy; test SMS delivery reliability; collect feedback on user experience. (4) Weeks 9-12: Document initial findings; finalize prototype (Phase 1 completion); prepare for Phase 2 scale-up planning; initiate funding applications (TIA, DSI); share code to GitHub and plan SADiLaR contribution. Budget requirement for Phase 1: R0 (using existing lab computers, free open-source tools, and SMS sandbox/developer tiers); Timeline contingency: R500 for unexpected costs